constants exactly. If we knew more about the interior nuclear forces, and about nuclear forces in particular, we could calculate them theoretically. Actually, however, only the converse method is available to us—their empirical determination on the ground of the data already known to us about the binding energies of nuclei, in other words, about mass defects. For instance, a study by Fliigge and v. Droste has resulted, by this method, in the following numerical values: A = 0.01574 a.m.u. ; B = 0.022 a.m.u.; C = 0.0165 a.m.u.; $\frac{3e^2}{5r_0}$ = 0.000646 a.m.u.

Thus the binding energies of the nuclei can actually be represented in good agreement with empirical findings. The curve representing our equation, shown in Figure 11, agrees approximately with the above-mentioned values of the constants; this curve shows the energy of the most stable elements for the atomic weights indicated. Since the binding energy is a negative quantity, the lower a point is, the greater is the binding energy of the element represented. The dots appearing along the curve are the values of binding energy per particle actually calculated from the mass defects. It will be seen that the agreement is very satisfactory.

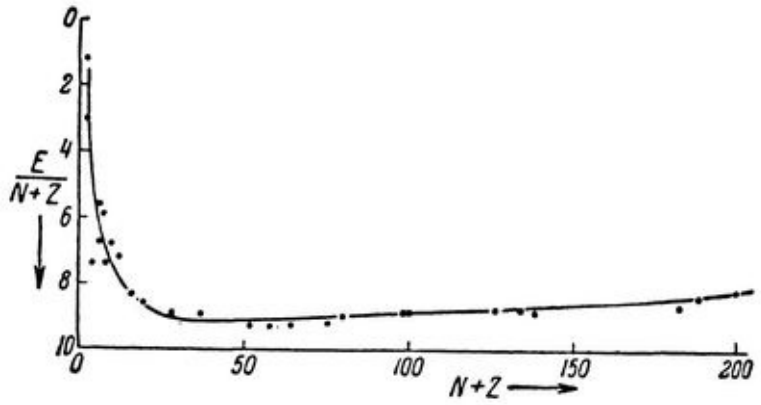


Figure 11.—Binding energies of nuclei as functions of $N + Z$.

Let us now analyse this curve. The position of its lowest point is certainly determined to a dominant extent by the largest term in our equation, $-A$. The rise observed for small atomic weights is due to the surface tension, which naturally plays the most important part for light nuclei. The rise for heavy nuclei is due to the electric repulsion of the protons.

Furthermore, our equation indicates that the lighter nuclei mostly contain approximately as many protons as neutrons. The last term, which is due to the force of repulsion, increases with the increase in the value of Z, and plays no important part as yet in the cases of light atoms, where Z is small. In such cases, it may therefore be disregarded. The third term is determined solely by the sum total of the number of particles, and not by their ratio, as in the